

BARDUFOSS, NorwayWMO: **010230**Lat: **69.056N**Lon: **18.540E**Elev: **252**StdP: **14.56**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.2	-11.2	-20.2	1.8	-15.0	-15.8	2.4	-10.8	24.5	25.3	21.7	28.2	1.8	290	0.700

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	75.2	59.5	71.3	58.1	67.6	56.5	61.7	71.1	59.5	68.0	57.7	65.5	6.5	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.6	71.5	65.4	55.6	66.4	62.7	53.8	62.1	60.8	27.7	70.8	26.2	68.1	25.0	65.7	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.0	13.7	-19.0	80.3	4.6	4.0	-22.3	83.2	-25.0	85.5	-27.6	87.7	-31.0	90.6		
(5)			-19.3	64.6	4.5	2.3	-22.6	66.2	-25.3	67.6	-27.8	68.9	-31.1	70.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.7	15.3	16.7	23.1	32.9	42.5	50.7	56.6	53.7	46.0	34.4	23.5	19.3
	DBStd	17.16	13.71	12.76	10.07	7.12	6.48	6.24	6.00	5.43	6.14	8.44	11.00	13.15
	HDD50	6080	1076	932	833	513	248	65	9	22	150	486	794	952
	HDD65	11083	1541	1352	1298	963	697	430	269	351	571	950	1244	1417
	CDD50	482	0	0	0	0	15	86	213	138	29	1	0	0
	CDD65	11	0	0	0	0	0	1	8	2	0	0	0	0
	CDH74	157	0	0	0	0	6	17	99	35	0	0	0	0
	CDH80	20	0	0	0	0	0	1	16	3	0	0	0	0
	WSAvg	4.9	4.3	4.8	5.4	5.7	6.2	6.1	5.3	4.5	4.4	4.1	3.8	4.4
	PrecAvg	34.1	3.4	2.9	2.9	1.8	1.7	1.9	2.8	2.7	3.4	3.9	2.8	3.7
	PrecMax	42.4	8.3	5.7	7.9	4.9	3.8	3.7	6.5	5.8	7.4	5.9	5.8	6.7
(15) Precipitation	PrecMin	26.7	0.4	0.4	0.7	0.3	0.4	0.2	0.7	0.8	1.2	1.4	0.1	0.6
	PrecStd	4.1	2.1	1.2	1.9	1.1	0.9	0.9	1.3	1.1	1.7	1.4	1.7	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	43.5	43.0	46.7	55.8	72.9	76.5	81.7	78.6	67.7	57.2	47.9	45.1	
	0.4% MCWB	38.6	38.3	38.1	44.6	54.9	57.9	64.6	60.1	55.8	49.1	43.6	40.5	
	2% DB	40.6	40.1	42.6	51.4	65.4	71.5	76.3	72.9	63.2	51.9	44.1	42.2	
	2% MCWB	36.8	36.3	36.8	42.0	51.0	56.0	60.8	59.6	54.2	46.3	40.5	38.3	
	5% DB	37.4	37.7	39.9	46.7	58.8	66.4	72.3	68.0	59.8	49.1	41.0	39.4	
	5% MCWB	34.7	34.8	35.3	39.7	47.2	53.5	59.3	57.9	52.5	45.0	37.8	35.7	
	10% DB	34.0	34.4	37.4	43.4	54.1	62.6	68.7	64.2	56.7	46.4	37.7	36.6	
	10% MCWB	31.9	31.8	33.4	37.5	44.8	51.9	57.9	55.9	50.7	42.8	34.9	33.7	
	0.4% WB	39.4	39.3	39.6	45.6	55.4	60.2	66.1	64.0	57.9	49.7	44.1	41.2	
	0.4% MCDB	42.6	41.8	44.3	54.0	71.6	72.0	79.3	72.8	64.5	54.4	46.8	43.8	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	36.9	36.7	37.5	42.8	51.1	57.3	63.0	60.7	55.3	47.5	41.0	38.6	
	2% MCDB	39.7	39.6	41.5	49.7	63.3	68.5	73.0	69.1	61.5	51.1	43.5	41.9	
	5% WB	34.6	34.9	35.8	40.3	48.0	55.1	60.7	58.7	53.2	45.2	37.9	36.1	
	5% MCDB	37.0	37.5	39.4	45.8	57.3	64.9	69.4	66.1	58.6	48.4	40.6	39.0	
	10% WB	32.0	32.1	33.7	38.2	45.4	52.9	58.7	56.9	51.3	43.0	35.2	33.7	
	10% MCDB	33.9	34.1	36.9	42.8	52.8	60.8	67.0	63.7	56.0	46.1	37.4	35.9	
(35) Mean Daily Temperature Range	MDBR	14.4	15.3	17.1	15.2	13.9	12.7	14.3	14.9	14.3	11.5	12.3	14.0	
	5% DB MCDBR	13.5	12.9	13.0	18.9	24.2	20.9	22.9	23.7	20.7	13.4	13.6	13.4	
	5% DB MCWBR	11.1	10.8	9.4	11.5	12.6	9.6	10.6	12.5	12.6	9.4	10.6	10.5	
	5% WB MCDBR	12.8	12.6	12.0	17.0	22.8	18.6	20.5	20.5	17.8	12.6	12.7	12.7	
	5% WB MCWBR	10.9	10.9	9.2	10.8	12.3	9.2	10.4	11.5	11.5	9.5	10.3	10.4	
(40) Clear-Sky Solar Irradiance	taub	0.210	0.263	0.309	0.331	0.343	0.344	0.349	0.336	0.314	0.274	0.361	N/A	
	taud	1.914	2.054	2.099	2.183	2.318	2.441	2.474	2.494	2.521	2.210	2.634	N/A	
	Ebn at Noon	10	150	218	254	267	271	265	255	231	149	21	0	
	Edh at Noon	2	14	26	33	33	30	28	24	18	12	3	0	
	RadAvg	6	104	491	1094	1514	1533	1405	1007	529	166	20	0	
(45) All-Sky Solar Radiation	RadStd	1	18	56	81	99	175	139	86	53	25	2	0	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A